

Neuro Lab: Measurement Station 2 – Notes

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Task 1: Series Resistor Measurement

$R_s = 2.17 \text{ k}\Omega$ (multimeter measurement), R_{smd} nominal: $4.701 \text{ k}\Omega$. Assembled circuit with R_s in series with R_{smd} . R_{smd} measured: $4.663 \text{ k}\Omega$. Value close to nominal, circuit validated. Code and circuit functioning.

Visuals

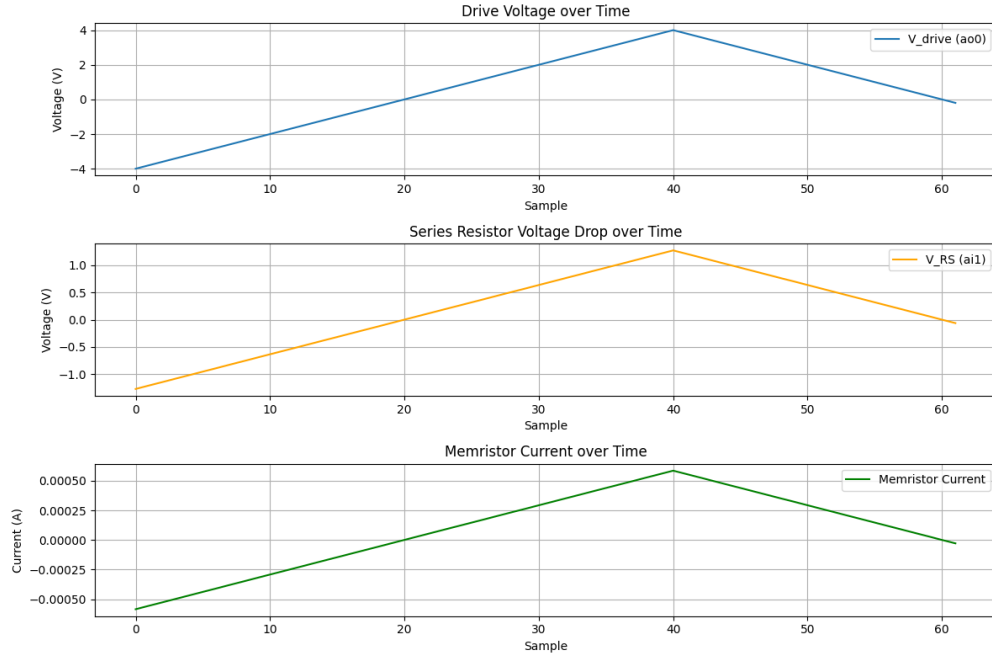


Figure 1: Task 1: Time Series Plot

Task 2: I(V) Curve Measurement

Observed I(V) characteristics with expected behavior; differential slope indicates evolving conduction. Early points show lower differential resistance consistent with onset of conduction increase.

Slope dI/dV_{bias} (first 5): [0.03837007, 0.05557502, 0.05442573, 0.02578657, 0.02225945] A/V

Differential resistance dV_{bias}/dI (first 5): [26.06198114, 17.99369474, 18.37366398, 38.77987233, 44.92473767] Ω

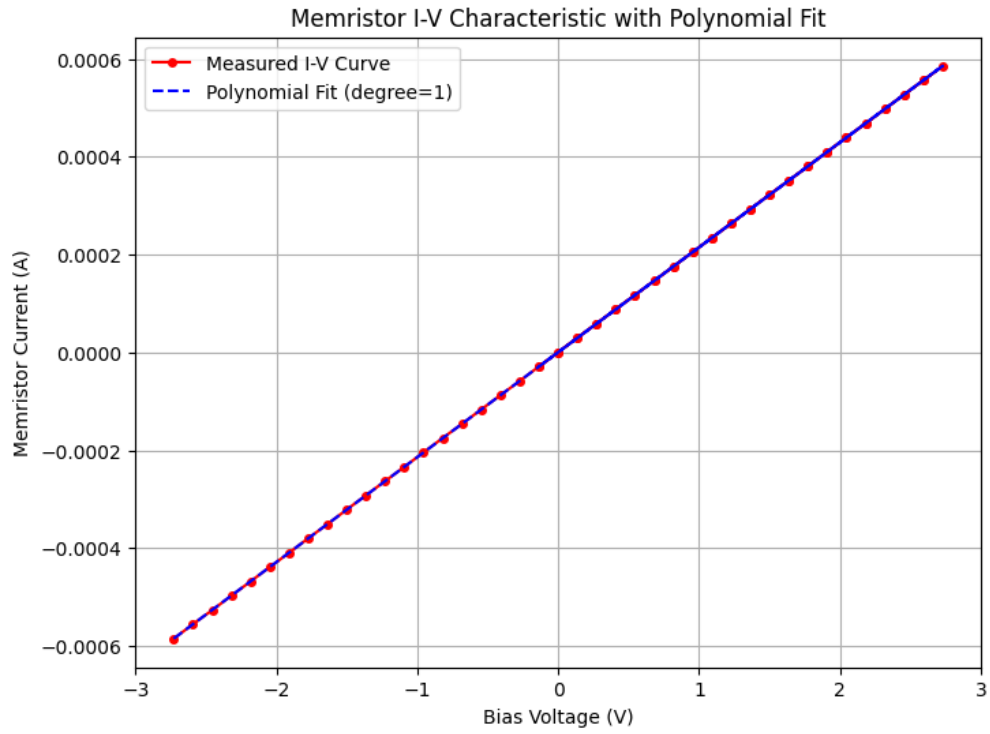


Figure 2: Task 1: Fit Plot

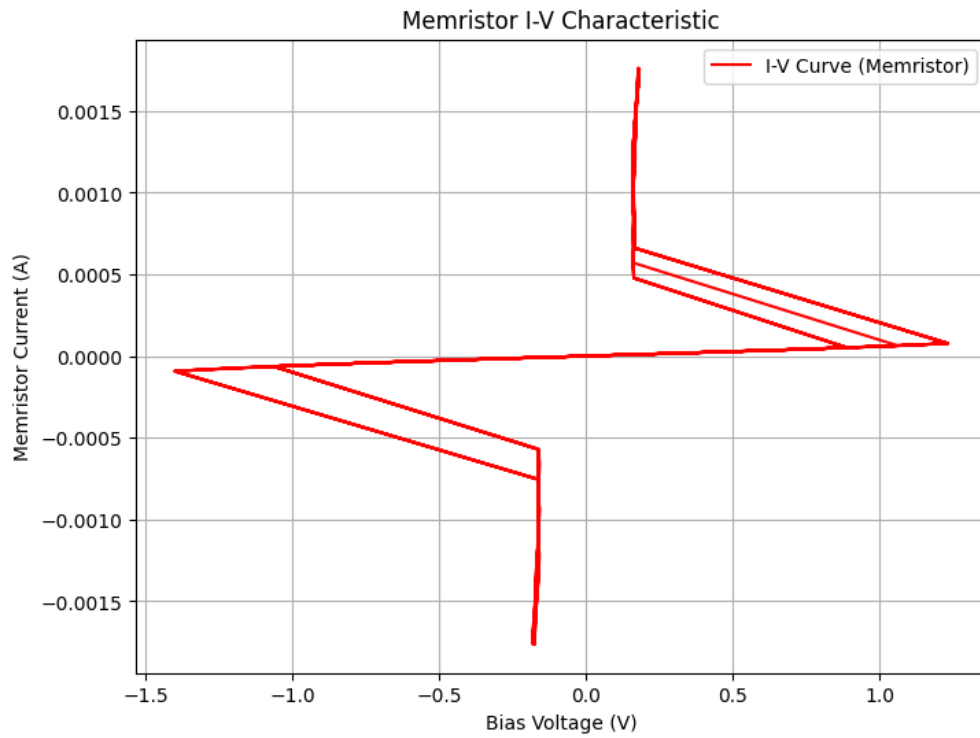


Figure 3: Task 2: Memristor I(V) Curve

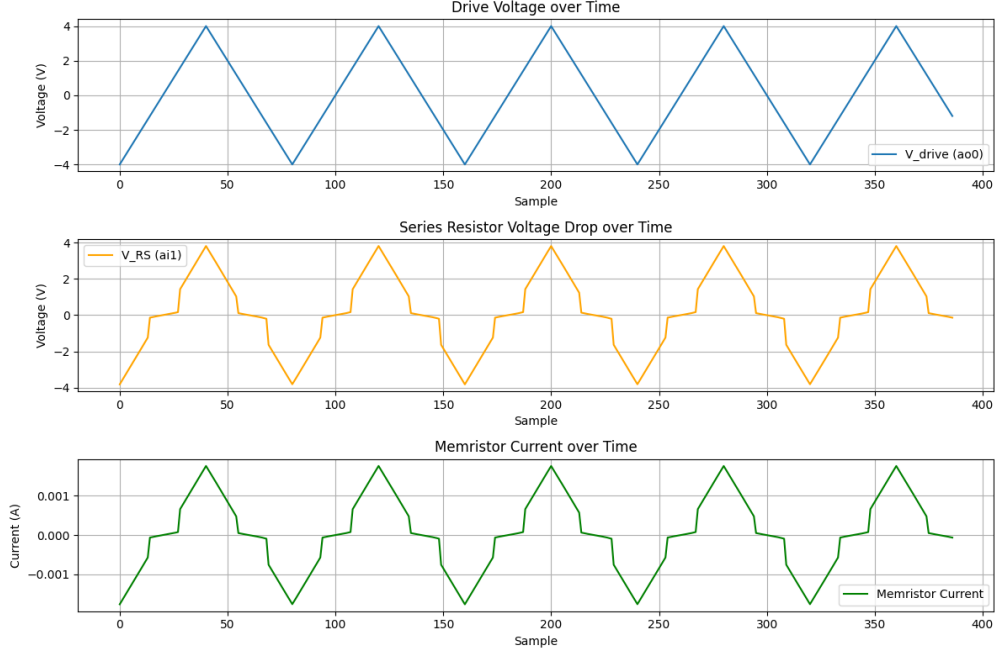


Figure 4: Task 2: Additional Plots

Visuals

Task 3: $R(T)$ Characteristics

$R(T)$ measurement implemented with VISA/SCPI control for the Peltier supply, DAQ-based voltage/current readout, and PT1000 temperature conversion. One run was noisy after moving the setup; after redoing with stable wiring and contact, a clean heating-cooling hysteresis was observed and transition temperature aligned with expectations.

Task 4: Relaxation (Reset) Time

Devices: VO_2 memristors at row 4, col 2 and row 4, col 3.

Method: Two-channel oscilloscope capture with $50\ \Omega$ termination; $V_{IN}(t)$ from AWG; $V_{TRANS}(t)$ at the oscilloscope input. Compute $R_M(t)$ via $R_M(t) = 2Z_0 \left(\frac{V_{IN}(t)}{V_{TRANS}(t)} - 1 \right)$ with $Z_0 = 50\ \Omega$. Clear LRS→HRS relaxation observed; reset time in the expected μs regime under our pulse/readout settings.

Future analysis: percentile-based time markers (e.g., 10–90% of ΔR) and stretched-exponential fits $R_M(t) = R_\infty - \Delta R e^{-(t/\tau)^\beta}$ to extract τ and β vs. pulse amplitude and readout level.

Brief Error Notes

Timing uncertainty from scope sampling/trigger; amplitude scale from $50\ \Omega$ tolerance and vertical calibration; residual reflections minimized with short cables and proper termination; thermal/mechanical stability improved repeatability.